Industrial Internship Report on

Crop and Weed Detection using YOLOv8

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was **Crop and Weed Detection using YOLOv8**, addresses one of the significant challenges in modern agriculture—accurately identifying weeds among crops to support precision farming. Traditional methods of weed identification rely heavily on manual inspection or indiscriminate pesticide spraying, both of which increase labor costs and chemical usage. The proposed solution uses the YOLOv8 object detection model to automatically detect crops and weeds from field images, enabling targeted weed management. This internship gave me a very good opportunity to get exposure to Industrial problems and design/ implement solution for that. It was an overall great experience to have this internship.

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1. Preface

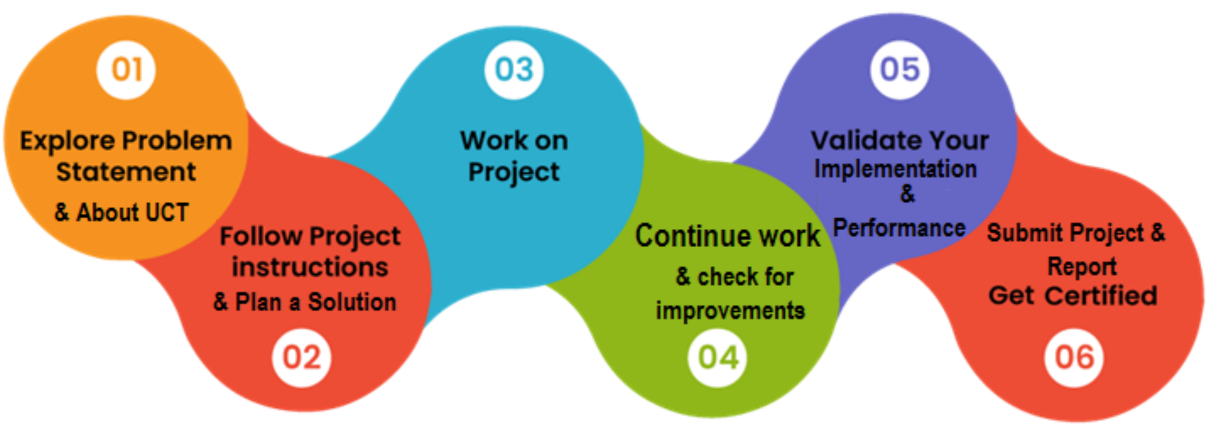
Industrial internships bridge the gap between academic learning and real-world engineering practice. They allow students to apply theoretical concepts to practical problems while gaining exposure to industry workflows, project management, and technical documentation.

This internship was conducted through **upskill Campus** in collaboration with **UniConverge Technologies Pvt. Ltd. (UCT)**, providing an opportunity to work on an AI-based computer vision project. The project selected for this internship was **Crop and Weed Detection using YOLOv8**, which aims to support precision agriculture through intelligent object detection.

Over the course of the internship, the project progressed through multiple stages, including understanding the problem statement, studying object detection techniques, preparing the dataset, planning the implementation, and documenting the proposed solution. The internship also provided an opportunity to strengthen technical skills in Python programming, deep learning, computer vision, and project documentation.

I sincerely thank **UniConverge Technologies Pvt. Ltd. (UCT)**, **upskill Campus**, the internship mentors, and my college for providing this valuable learning opportunity. Their guidance and support played an important role in enhancing my technical knowledge and practical understanding of AI-based applications. I also thank my family and friends for their encouragement throughout the internship.

Finally, I hope this report serves as a clear record of the knowledge gained and the work carried out during the internship, while also providing a strong foundation for future learning and professional growth.



2. Introduction

2.1. About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

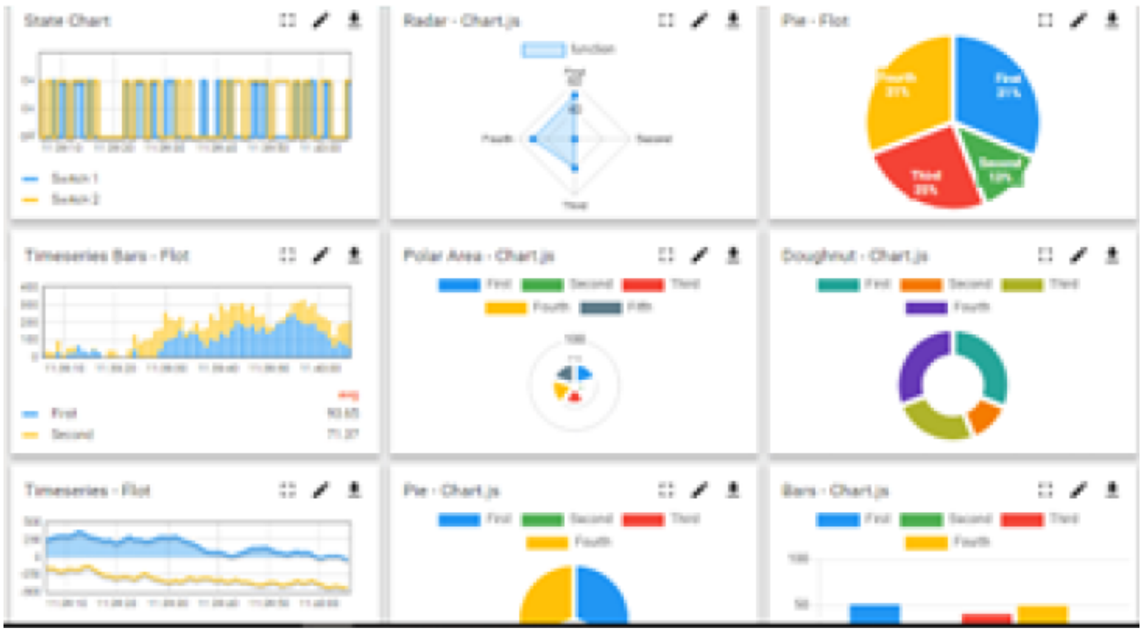
i. UCT IoT Platform (uct Insight)

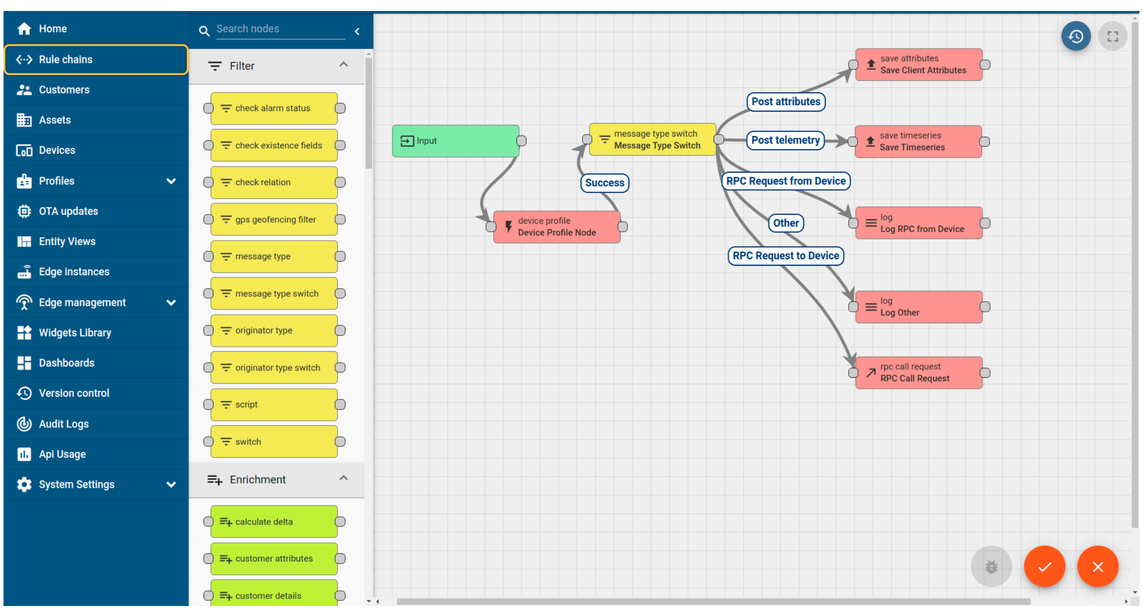
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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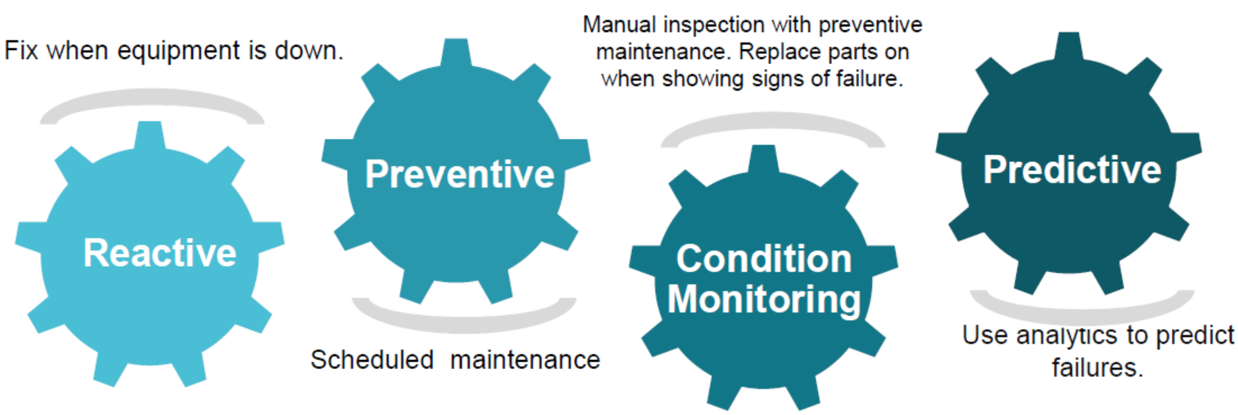
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

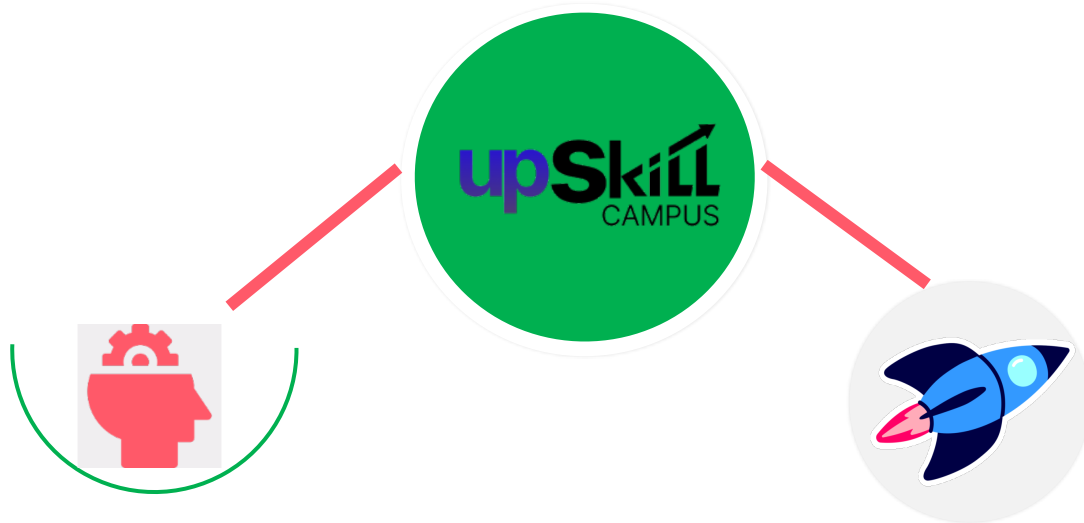
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2. About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

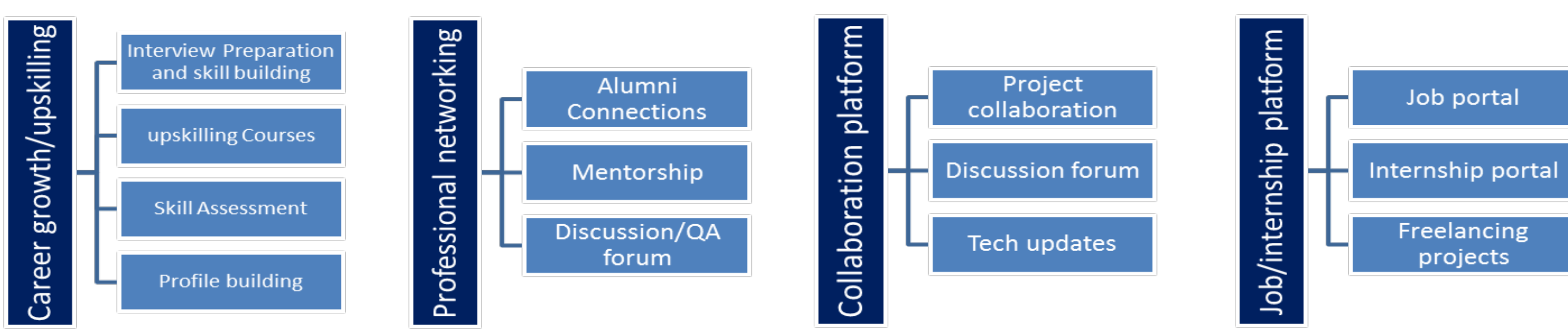
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable wa



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3. The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4. Objectives of this Internship program

The objective for this internship program was to

☛ get practical experience of working in the industry.

☛ to solve real world problems.

☛ to have improved job prospects.

☛ to have Improved understanding of our field and its applications.

☛ to have Personal growth like better communication and problem solving.

2.5. Reference

The following resources were referred to during the internship:

1. Ultralytics YOLOv8 Official Documentation.
2. OpenCV Official Documentation.
3. Python Official Documentation.
4. Research papers related to Crop and Weed Detection using Deep Learning.
5. Dataset documentation and annotation guidelines.
6. UniConverge Technologies Pvt. Ltd. internship materials.

2.6. Glossary

Terms	Acronym
YOLO	You Only Look Once
IoU	Intersection over Union
mAP	Mean Average Precision
CNN	Convolutional Neural Network
GPU	Graphics Processing Unit

3. Problem Statement

Agriculture is one of the most important sectors contributing to food production and economic development. However, one of the major challenges faced by farmers is the uncontrolled growth of weeds in crop fields. Weeds compete with crops for essential resources such as water, nutrients, sunlight, and space, leading to reduced crop yield and lower agricultural productivity.

The traditional method of weed control involves manual identification followed by uniform pesticide spraying across the entire field. Manual inspection is time-consuming, labor-intensive, and often impractical for large-scale farms. Similarly, spraying pesticides over the entire field results in excessive chemical usage, increased production costs, environmental pollution, and possible contamination of healthy crops.

To overcome these challenges, there is a need for an intelligent system capable of accurately distinguishing crops from weeds. Recent advancements in Artificial Intelligence and Computer Vision have enabled automated object detection techniques that can identify different plant species directly from images.

The objective of this project is to develop a **Crop and Weed Detection System using YOLOv8**. The proposed system detects crops and weeds in agricultural field images by generating bounding boxes around the detected objects. Such a system can assist precision agriculture by enabling selective pesticide spraying, reducing chemical wastage, minimizing crop damage, and improving agricultural productivity.

This project demonstrates how deep learning can be effectively applied to solve practical agricultural problems while promoting sustainable farming practices.

4. Existing and Proposed solution

Existing Solution

Several traditional methods are currently used for weed management in agriculture.

Manual Weed Identification

Farmers visually inspect fields to identify weeds before removing them manually or applying pesticides.

Limitations

- Time-consuming
- Labor-intensive
- Difficult for large agricultural fields
- Human errors in weed identification

Uniform Pesticide Spraying

Pesticides are sprayed throughout the entire field without distinguishing crops from weeds.

Limitations

- Excessive pesticide consumption
- Increased farming cost
- Chemical contamination of crops
- Environmental pollution
- Reduced soil quality over time

Although these methods are widely practiced, they are inefficient and do not support precision agriculture.

Proposed Solution

The proposed solution uses **YOLOv8 (You Only Look Once Version 8)**, a state-of-the-art deep learning object detection algorithm, to automatically identify crops and weeds from agricultural images.

The workflow consists of:

- Collecting agricultural field images.
- Preprocessing the dataset.
- Annotating images using the YOLO format.

- Training a custom YOLOv8 model.
- Detecting crops and weeds in unseen images.
- Displaying bounding boxes with confidence scores.

Instead of spraying pesticides across the entire field, the detection results can be integrated with smart spraying systems to target only weeds.

Advantages

- Faster detection
- Reduced pesticide usage
- Higher agricultural productivity
- Lower operational cost
- Environment-friendly farming
- Supports precision agriculture

The proposed Crop and Weed Detection System follows a structured workflow consisting of multiple stages. Each stage contributes to improving the accuracy and efficiency of the object detection model.

4.1. Code submission (Github link)

https://github.com/swethasivakumaran18-creator/upskillcampus

4.2. Report submission (Github link) :

https://github.com/swethasivakumaran18-creator/upskillcampus/blob/main/InternshipReport.pdf

5. Proposed Design/ Model

The proposed Crop and Weed Detection System follows a structured workflow consisting of multiple stages. Each stage contributes to improving the accuracy and efficiency of the object detection model.

Dataset Collection

The first stage involves collecting images of agricultural fields containing both crops and weeds.

Initially,

- 589 field images were collected.

Each image contained different lighting conditions, plant densities, and background variations to improve model generalization.

Dataset Cleaning

The collected images were carefully reviewed.

Images that were

- blurred,
- duplicated,
- poorly illuminated,
- or contained insufficient information

were removed.

After cleaning,

546 high-quality images remained for further processing.

Image Preprocessing

The original image resolution was approximately

4000 × 3000 pixels

Training directly on such large images would significantly increase computational requirements.

Therefore, all images were resized to

512 × 512 × 3

while preserving the visual information required for detection.

Data Augmentation

Since 546 images were insufficient for training a deep learning model, various augmentation techniques were applied.

These include

- Horizontal Flip
- Vertical Flip
- Rotation
- Brightness Adjustment
- Zoom
- Translation

Using augmentation,

the dataset size increased to

1300 images.

This helped reduce overfitting and improved the model's robustness.

Image Annotation

Each image was manually annotated using the YOLO annotation format.

Bounding boxes were drawn around

- Crop
- Weed

Every annotation file stores

- object class
- bounding box coordinates

These annotations are used by YOLOv8 during training.

Model Training

The prepared dataset is used to train the YOLOv8 object detection model.

The model learns

- object localization
- feature extraction
- classification

through multiple epochs using transfer learning.

Training parameters include

- Image Size: **512 × 512**
- Batch Size: **16**
- Epochs: **50**
- Framework: **Ultralytics YOLOv8**

Object Detection

After training,

the model predicts

- Crop locations
- Weed locations

using

- Bounding Boxes
- Confidence Scores
- Class Labels

The final output enables automatic identification of weeds, which can support selective pesticide spraying.

Performance evaluation is an important stage in any industrial AI application. It helps determine whether the developed system meets the desired objectives and identifies areas for improvement. For the Crop and Weed Detection project, the performance evaluation focuses on the model's ability to accurately detect crops and weeds from agricultural images.

The YOLOv8 model will be evaluated using standard object detection metrics such as Precision, Recall, Mean Average Precision (mAP), and inference time. These metrics help measure detection accuracy, localization performance, and prediction speed.

5.1. Test Plan/ Test Cases

The following test cases are planned to evaluate the system.

Test Case	Objective	Expected Result
TC-01	Load the dataset	Dataset loads successfully
TC-02	Train the model	Model completes training without errors
TC-03	Validate the model	Performance metrics are generated
TC-04	Test on unseen images	Crops and weeds are detected correctly
TC-05	Display predictions	Bounding boxes and confidence scores are displayed

5.2. Test Procedure

The following steps will be used to evaluate the system:

1. Load the prepared dataset.
2. Configure the YOLOv8 model.
3. Train the model using the training dataset.
4. Validate the trained model using the validation dataset.
5. Test the model on previously unseen agricultural images.
6. Compare prediction results with ground-truth annotations.
7. Record performance metrics and analyze the results.

5.3. Performance Outcome

The final report should include:

- Precision
- Recall
- Mean Average Precision (mAP)
- Training Loss
- Validation Loss

- Sample prediction images
- Confusion Matrix (if generated)
- Training graphs (loss and metric curves)

These results will demonstrate the effectiveness of the proposed system and help evaluate its suitability for precision agriculture.

6. My learnings

The internship provided valuable practical exposure to Artificial Intelligence, Machine Learning, and Computer Vision. Through this project, I gained a deeper understanding of how object detection techniques can be applied to solve real-world agricultural problems.

Key learnings include:

- Understanding the complete workflow of an AI-based object detection project.
- Learning the fundamentals of YOLOv8 architecture and transfer learning.
- Gaining experience in dataset preparation, image preprocessing, and annotation.
- Understanding the importance of data quality in machine learning projects.
- Learning to configure, train, and evaluate deep learning models.
- Improving Python programming skills for AI applications.
- Developing technical documentation and project management skills.
- Understanding the use of GitHub for maintaining project code and documentation.
- Enhancing analytical thinking and problem-solving abilities through practical implementation.

Overall, the internship strengthened my confidence in applying theoretical concepts to practical applications and improved my readiness for future opportunities in Data Science and Artificial Intelligence.

7. Future work scope

Although the proposed Crop and Weed Detection system provides an effective approach for identifying crops and weeds, several improvements can be incorporated in future versions.

Possible future enhancements include:

- Deploying the trained model on embedded devices such as Raspberry Pi or NVIDIA Jetson for real-time field applications.
- Integrating the detection system with automated pesticide spraying mechanisms.
- Expanding the dataset to include additional crop and weed species.

- Improving model accuracy using larger datasets and advanced YOLO architectures.
- Developing a mobile application that allows farmers to perform weed detection directly from smartphones.
- Integrating drone-based image capture for large-scale agricultural monitoring.
- Incorporating real-time GPS-based weed mapping for precision farming.
- Optimizing the system for faster inference and lower computational requirements.

These enhancements can further improve the practical usability of the system and contribute to sustainable agricultural practices.